Nonlinear scalar scattering on Kerr from an endpoint spectral Green-function method

KerrScattering Project[1]

[1] *Manuscript draft generated from the KerrScattering repository*

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We formulate and compute the leading weakly nonlinear scattering response of a self-interacting complex scalar field on Kerr. The calculation is the rotating black-hole continuation of a Schwarzschild nonlinear Green-function spectral method. The scalar Teukolsky equation is solved in the compact coordinate $z = r_+/r$, with infinity and the future horizon retained as collocation endpoints. After peeling the known phases, the degenerate endpoint rows of the radial equation provide regular first-order boundary conditions. The linear horizon-normalized in mode is matched to down/up solutions at an intermediate radius to obtain $B^{\rm inc}$ and $B^{\rm ref}$. Against 37 scalar GeneralizedSasakiNakamura.jl benchmark cases, the worst invariant amplitude error is 1.485×10^{-8} and the worst flux residual is 3.714×10^{-9}. For a cubic interaction $\Box\Phi + \epsilon|\Phi|^2\Phi = 0$, the Kerr factor $\Sigma = r^2 + a^2\cos^2\theta$ generates a projected radial source $(r^2 C^{(0)}_{\ell'\ell m} + a^2 C^{(2)}_{\ell'\ell m})|R_{\ell m}|^2 R_{\ell m}$. We compute the first-order Green-function amplitudes for self and target channels, including a representative $\ell' = 2, \ldots, 6$ channel scan for $(a, \ell, m, M\omega) = (0.5, 2, 2, 0.30)$. For the self channel, converting the unit-horizon response to fixed incident normalization gives $T^{(1)} + R^{(1)} \simeq 0$, the Kerr analogue of the nonlinear balance law in the Schwarzschild problem. Off-diagonal target channels are nonzero at the field-amplitude level but enter energy flux only at quadratic order in the nonlinear correction.

I. INTRODUCTION

Scattering by black holes provides observables complementary to quasinormal mode spectra. In linear theory, scalar, electromagnetic, and gravitational waves on Schwarzschild and Kerr backgrounds exhibit barrier penetration, greybody factors, superradiance, and resonance structures which are well understood through partial waves and related complex-angular-momentum methods [1–7]. These quantities are also relevant for recent frequency-domain descriptions of black-hole ringdowns in terms of greybody factors rather than only sums of quasinormal modes [8, 9].

The nonlinear scattering problem is less developed. The Schwarzschild calculation used as the template for the present work considers a weak self-interaction and computes the first nonlinear correction by feeding the linear in solution into a Green-function source. Its central structure is simple: solve the homogeneous scattering problem accurately, project the nonlinear source onto partial waves, impose no incoming nonlinear radiation, and read off the first-order correction to the horizon and infinity fluxes.

Here we carry this strategy to the scalar Teukolsky equation on Kerr, keeping the same computational philosophy as the Schwarzschild spectral code. The new features are genuinely Kerr-specific. First, the angular basis is spheroidal, not spherical. Second, the radial equation contains the horizon frequency $p = \omega - m\Omega_H$, so the linear flux law allows superradiant reflection. Third, the nonlinear source is not merely a spherical Gaunt coefficient times a radial power: multiplying the equation by $\Sigma = r^2 + a^2\cos^2\theta$ produces two angular projectors and couples the source mode (ℓ, m) to target channels ℓ' of the same m.

The paper follows the structure of the Schwarzschild nonlinear Green-function template. Section II defines the nonlinear Kerr scalar scattering problem. Section III describes the endpoint spectral method and the nonlinear Green-function amplitudes. Section IV reports the linear validation, superradiant frequency sweep, nonlinear self-channel balance, and the first target-channel matrix result. The appendices give the working derivations used by the code so that normalization and sign conventions can be checked without referring to a separate note.

II. NONLINEAR SCATTERING PROBLEM ON KERR

A. Scalar Teukolsky equation

In Boyer–Lindquist coordinates,

$$\Sigma = r^2 + a^2\cos^2\theta, \tag{1}$$

$$\Delta = r^2 - 2Mr + a^2 = (r - r_+)(r - r_-), \tag{2}$$

$$r_+ = M + \sqrt{M^2 - a^2}, \qquad r_- = M - \sqrt{M^2 - a^2}, \tag{3}$$

and

$$\Omega_H = \frac{a}{r_+^2 + a^2}. \tag{4}$$

We use the Kerr tortoise coordinate

$$\frac{\mathrm{d}r_*}{\mathrm{d}r} = \frac{r^2 + a^2}{\Delta}. \tag{5}$$

The additive constant in r_* fixes only an overall scattering phase.

For the scalar mode

$$\Phi^{(0)}_{\ell m\omega} = R_{\ell m\omega}(r) S_{\ell m}(\theta; a\omega) e^{\mathrm{i}m\phi - \mathrm{i}\omega t}, \tag{6}$$

the scalar wave operator separates after multiplying by Σ. Substitution gives the identity

$$0 = \frac{1}{R}\frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta\frac{\mathrm{d}R}{\mathrm{d}r}\right) + \frac{K^2}{\Delta} - a^2\omega^2 + 2am\omega + \frac{1}{S\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}S}{\mathrm{d}\theta}\right) + a^2\omega^2\cos^2\theta - \frac{m^2}{\sin^2\theta}. \quad (7)$$

The r-dependent and θ-dependent parts can therefore be set equal to a single separation constant $A_{\ell m}$. With the convention used here, the angular equation is

$$0 = \frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}S_{\ell m}}{\mathrm{d}\theta}\right) + \left[a^2\omega^2\cos^2\theta - \frac{m^2}{\sin^2\theta} + A_{\ell m}\right]S_{\ell m}, \quad (8)$$

with unit normalization on the sphere. The radial equation is

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta\frac{\mathrm{d}R_{\ell m\omega}}{\mathrm{d}r}\right) + \left(\frac{K^2}{\Delta} - \lambda_{\ell m\omega}\right)R_{\ell m\omega} = 0, \quad (9)$$

where

$$K = (r^2+a^2)\omega - am, \qquad \lambda_{\ell m\omega} = A_{\ell m} + a^2\omega^2 - 2am\omega. \quad (10)$$

For $a\omega = 0$, $S_{\ell m}$ reduces to the spherical harmonic and $\lambda = \ell(\ell+1)$.

B. Linear boundary conditions and fluxes

The horizon-normalized in solution is defined by

$$R^{\rm in}_{\ell m\omega} \sim e^{-\mathrm{i}pr_*}, \qquad p = \omega - m\Omega_H, \qquad r \to r_+, \quad (11)$$

and at infinity by

$$R^{\rm in}_{\ell m\omega} \sim B^{\rm inc}_{\ell m\omega}\frac{e^{-\mathrm{i}\omega r_*}}{r} + B^{\rm ref}_{\ell m\omega}\frac{e^{+\mathrm{i}\omega r_*}}{r}. \quad (12)$$

With this convention the incident amplitude is $B^{\rm inc}$, the reflected amplitude is $B^{\rm ref}$, and the horizon transmission amplitude is unity. The invariant linear reflection and transmission factors for a unit incident wave are

$$R^{(0)} = \left|\frac{B^{\rm ref}}{B^{\rm inc}}\right|^2, \qquad T^{(0)} = \frac{p(r_+^2+a^2)}{\omega|B^{\rm inc}|^2}, \quad (13)$$

so that

$$T^{(0)} + R^{(0)} = 1. \quad (14)$$

This follows directly from the conserved radial current $J = (2\mathrm{i})^{-1}\Delta(R^*R' - RR^{*\prime})$: the unit-horizon in mode gives $J_H = -p(r_+^2+a^2)$ at the future horizon and $J_\infty = -\omega|B^{\rm inc}|^2 + \omega|B^{\rm ref}|^2$ at infinity. Hence

$$\omega\left(|B^{\rm inc}|^2 - |B^{\rm ref}|^2\right) = p(r_+^2+a^2), \quad (15)$$

which is Eq. (13). More details are collected in Appendix A. When $p<0$, $T^{(0)}<0$ and the reflected flux exceeds the incident flux: this is scalar superradiance.

C. Weak cubic source

We take the self-interacting scalar equation to be

$$\Box\Phi + \epsilon|\Phi|^2\Phi = 0, \qquad |\epsilon| \ll 1, \quad (16)$$

and expand

$$\Phi = \Phi^{(0)} + \epsilon\Phi^{(1)} + \mathcal{O}(\epsilon^2). \quad (17)$$

For a monochromatic source mode (ℓ, m, ω), the cubic term has the same time and azimuthal dependence. We therefore write

$$\Phi^{(1)} = e^{\mathrm{i}m\phi - \mathrm{i}\omega t}\sum_{\ell'\ge|m|} R^{(1)}_{\ell'|\ell m\omega}(r)S_{\ell' m}(\theta; a\omega). \quad (18)$$

Multiplying the scalar wave equation by Σ and projecting onto the target spheroidal harmonic gives

$$\mathcal{L}_{\ell'}R^{(1)}_{\ell'|\ell m\omega} = -Q_{\ell'\ell m}(r), \quad (19)$$

where $\mathcal{L}_{\ell'}$ denotes the radial operator in Eq. (9) with $\lambda_{\ell' m\omega}$, and

$$Q_{\ell'\ell m}(r) = \left[r^2C^{(0)}_{\ell'\ell m} + a^2C^{(2)}_{\ell'\ell m}\right]|R^{\rm in}_{\ell m\omega}|^2R^{\rm in}_{\ell m\omega}. \quad (20)$$

The two angular projectors are

$$C^{(s)}_{\ell'\ell m} = \int \mathrm{d}\Omega\,\cos^s\theta\,S^*_{\ell' m}|S_{\ell m}|^2S_{\ell m}, \qquad s = 0, 2. \quad (21)$$

The factor $r^2C^{(0)} + a^2C^{(2)}$ is simply the decomposition of $\Sigma|S_{\ell m}|^2S_{\ell m}$ into the target basis: the r^2 piece carries the ordinary spheroidal Gaunt coefficient, while the $a^2\cos^2\theta$ piece is a genuinely Kerr correction. This is the first structural difference from the Schwarzschild calculation: even a single incident spheroidal mode sources a target-channel matrix.

III. METHODS

A. Endpoint pseudo-spectral homogeneous solutions

We solve Eq. (9) on two Chebyshev subdomains in

$$z = \frac{r_+}{r}, \qquad z = 0 \leftrightarrow r = \infty, \qquad z = 1 \leftrightarrow r = r_+. \quad (22)$$

The radial derivatives are converted by

$$\frac{\mathrm{d}}{\mathrm{d}r} = -\frac{z^2}{r_+}\frac{\mathrm{d}}{\mathrm{d}z}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}r^2} = \frac{z^4}{r_+^2}\frac{\mathrm{d}^2}{\mathrm{d}z^2} + \frac{2z^3}{r_+^2}\frac{\mathrm{d}}{\mathrm{d}z}. \quad (23)$$

The known phases are peeled off before discretization,

$$R_{\rm in} = e^{-\mathrm{i}pr_*}u_{\rm in}(z), \quad (24)$$

$$R_{\rm down} = \frac{e^{-\mathrm{i}\omega r_*}}{r}u_{\rm down}(z), \quad (25)$$

$$R_{\rm up} = \frac{e^{+\mathrm{i}\omega r_*}}{r}u_{\rm up}(z). \quad (26)$$

The transformed equation has the schematic form

$$B_2(z)u''(z) + B_1(z)u'(z) + B_0(z)u(z) = 0. \tag{27}$$

At $z = 0$ for down/up modes and at $z = 1$ for the in mode, B_2 vanishes. Thus the endpoint row is a first-order regularity condition rather than an externally imposed truncation. The normalizations are

$$u_{\rm down}(0) = 1, \qquad u_{\rm up}(0) = 1, \qquad u_{\rm in}(1) = 1. \tag{28}$$

For low frequencies we use an analytic sinh mapping to cluster points toward the far-field endpoint, following the same motivation as the Schwarzschild Gevrey-regularity discussion in the base calculation.

At an intermediate point $z_m = r_+/r_m$, the horizon solution is matched to down/up solutions:

$$R_{\rm in} = B^{\rm inc} R_{\rm down} + B^{\rm ref} R_{\rm up}. \tag{29}$$

Matching both R and $\mathrm{d}R/\mathrm{d}r$ gives the two-by-two linear system

$$\begin{pmatrix} R_{\rm down} & R_{\rm up} \\ R'_{\rm down} & R'_{\rm up} \end{pmatrix}_{r_m} \begin{pmatrix} B^{\rm inc} \\ B^{\rm ref} \end{pmatrix} = \begin{pmatrix} R_{\rm in} \\ R'_{\rm in} \end{pmatrix}_{r_m}. \tag{30}$$

This is the same algebraic matching used in the Schwarzschild spectral route; only the phase factors and the radial operator have changed.

B. Green-function nonlinear amplitudes

Let $R^{\rm in}_{\ell' m\omega}$ and $R^{\rm up}_{\ell' m\omega}$ denote the target-channel in and up solutions. The radial Wronskian

$$\mathcal{W}_{\ell'} = \Delta \left(R^{\rm in}_{\ell'} \frac{\mathrm{d}R^{\rm up}_{\ell'}}{\mathrm{d}r} - R^{\rm up}_{\ell'} \frac{\mathrm{d}R^{\rm in}_{\ell'}}{\mathrm{d}r} \right) \tag{31}$$

is independent of r. The first-order solution is required to contain no incoming nonlinear radiation from infinity and no outgoing nonlinear radiation from the past horizon. The two Green-function amplitudes are therefore

$$A^{(1)}_{{\rm ref},\ell'|\ell m} = -\frac{1}{\mathcal{W}_{\ell'}} \int_{r_+}^{\infty} R^{\rm in}_{\ell'}(r) Q_{\ell'\ell m}(r)\,\mathrm{d}r, \tag{32}$$

$$A^{(1)}_{H,\ell'|\ell m} = -\frac{1}{\mathcal{W}_{\ell'}} \int_{r_+}^{\infty} R^{\rm up}_{\ell'}(r) Q_{\ell'\ell m}(r)\,\mathrm{d}r. \tag{33}$$

These are the quantities directly computed by the code in unit-horizon normalization.

The sign in Eqs. (32) and (33) comes from $\mathcal{L}R^{(1)} = -Q$. Equivalently, the Green function normalized by the jump condition $\Delta[\partial_r G]^{r=r'^-}_{r=r'^+} = 1$ is

$$G_{\ell'}(r, r') = \frac{R^{\rm in}_{\ell'}(r_<) R^{\rm up}_{\ell'}(r_>)}{\mathcal{W}_{\ell'}}, \tag{34}$$

and $R^{(1)} = -\int GQ\,\mathrm{d}r'$. Taking r to infinity picks out the coefficient of $R^{\rm up}$ and gives Eq. (32); taking r to the future horizon picks out the coefficient of $R^{\rm in}$ and gives Eq. (33).

The outer part of the integrals is highly oscillatory. On the outer domain, $R_{\rm in}$ is a sum of down and up phases, while $R_{\rm up}$ is outgoing. The products in Eqs. (32) and (33) decompose into phase channels $\exp(\mathrm{i}n\omega r_*)$ with $n = -4, -2, 0, 2, 4$. We integrate the exterior tail in r_* with Fourier-weighted quadrature, while the inner finite interval is integrated by Gauss-Legendre quadrature in z.

C. Fixed incident normalization and nonlinear balance

The amplitudes in Eqs. (32) and (33) correspond to a linear solution with unit horizon transmission. To compare with the Schwarzschild nonlinear scattering coefficients, we fix the incident amplitude to unity. The rescaled field is

$$R^{(0)}_{\rm phys} = \alpha R_{\rm in}, \qquad \alpha = \frac{1}{B^{\rm inc}}. \tag{35}$$

Since the source is cubic, the first-order field scales as $|\alpha|^2\alpha$. For the self channel $\ell' = \ell$, the physical reflected and horizon amplitudes are

$$\mathcal{R}_{\rm phys} = \frac{B^{\rm ref}}{B^{\rm inc}} + \epsilon \frac{A^{(1)}_{\rm ref}}{|B^{\rm inc}|^2 B^{\rm inc}}, \tag{36}$$

$$\mathcal{H}_{\rm phys} = \frac{1}{B^{\rm inc}} + \epsilon \frac{A^{(1)}_{H}}{|B^{\rm inc}|^2 B^{\rm inc}}. \tag{37}$$

The first nonlinear corrections to the flux coefficients are

$$R^{(1)} = \frac{2\,\mathrm{Re}\left[\left(B^{\rm ref}\right)^* A^{(1)}_{\rm ref}\right]}{|B^{\rm inc}|^4}, \tag{38}$$

$$T^{(1)} = \frac{2p(r_+^2 + a^2)}{\omega |B^{\rm inc}|^4} \mathrm{Re}\, A^{(1)}_{H}. \tag{39}$$

For a self-channel source, conservation implies

$$T^{(1)} + R^{(1)} = 0, \tag{40}$$

up to numerical error. In the present cubic model this is the first-order part of the conserved $U(1)$ Klein-Gordon flux: the source $Q \propto |R_{\rm in}|^2 R_{\rm in}$ gives a real overlap with $R^*_{\rm in}$, so it redistributes flux between horizon and infinity but does not create a net monochromatic current. Off-diagonal target channels are orthogonal to the linear outgoing channel in the angular integral. They are first-order field amplitudes, but their direct energy-flux contribution starts at $\mathcal{O}(\epsilon^2)$.

IV. RESULTS

A. Linear validation and superradiance

The linear solver was validated against Generalized-SasakiNakamura.jl in the unit-Teukolsky-transmission convention [10]. Complex phases were not used as pass/fail quantities because a shift of the Kerr tortoise coordinate changes them. The benchmark gates were imposed on $|B^{\rm inc}|$, $|B^{\rm ref}|$, and the flux balance.

B. Self-channel nonlinear flux coefficients

Table II gives the fixed-incident self-channel coefficients of Eqs. (38) and (39). The Schwarzschild row is included as a limit check against the base problem. For ordinary absorption ($p > 0$), a positive cubic coupling in the present sign convention increases the transmitted flux and decreases the reflected flux. In the superradiant regime ($p < 0$), $T^{(0)} < 0$ and the signs reverse in the same flux-balance sense.

The nonlinear balance residual $T^{(1)}+R^{(1)}$ is 3.3×10^{-12} for the Schwarzschild row, -5.6×10^{-13} for the absorbing Kerr $a = 0.5$ row, and -2.7×10^{-13} for the high-spin superradiant row. The $a = 0.5$, $M\omega = 0.24$ row is closer to the superradiant threshold and shows a larger absolute residual, -1.1×10^{-8}, because the two nonlinear flux terms are obtained from large nearly cancelling Green-function amplitudes.

C. Target-channel matrix

For the representative absorbing Kerr mode $(a, \ell, m, M\omega) = (0.5, 2, 2, 0.30)$, the cubic source was projected onto target channels $\ell' = 2, \ldots, 6$. Odd target channels vanish by angular parity in this case. The production computation used $N = 288$, $q = 224$, and Fourier-tail tolerance 10^{-11}.

A fixed-$q = 224$ convergence scan over $N = 224, 256, 288$ gives a largest reflected-amplitude change of 1.14×10^{-4} relative to the $N = 288$ reference row, while the change from $N = 256$ to $N = 288$ is at most 6.27×10^{-7} over the active reflected channels. The off-diagonal horizon amplitudes are more cancellation limited. The $\ell' = 4$ horizon amplitude is stable at roughly the percent level between $N = 256$ and $N = 288$, whereas $\ell' = 6$ is a near-zero 10^{-7} quantity and should be interpreted only as an order-of-magnitude diagnostic at current double precision.

TABLE I. Linear scalar validation summary. The benchmark grid contains Schwarzschild, moderate-spin, high-spin, superradiant, and non-superradiant cases.

quantity	value
usable GSN benchmark cases	37
frequency-sweep points	51
worst amplitude-magnitude error	1.485×10^{-8}
worst flux residual	3.714×10^{-9}
largest sampled superradiant gain	4.970×10^{-4}

V. DISCUSSION AND NEXT STEPS

The central conclusion is that the Schwarzschild nonlinear Green-function workflow extends cleanly to Kerr scalar scattering once the spheroidal angular projection and the Kerr flux normalization are handled explicitly. The linear piece is already benchmarked to the level required for a production solver. The nonlinear self-channel reproduces the expected first-order flux balance after converting from unit-horizon to fixed incident normalization. The Kerr geometry also exposes a new object absent from the simple Schwarzschild self-channel calculation: the target-channel amplitude matrix $A^{(1)}_{{\rm ref},\ell'|\ell m}$ and $A^{(1)}_{H,\ell'|\ell m}$.

The present results should be read with two normalization caveats. First, the tabulated Green amplitudes are not by themselves physical nonlinear corrections; the physical field correction is $\epsilon A^{(1)}$, and fixed incident normalization introduces the cubic scaling in Eqs. (38)–(39). Second, off-diagonal first-order fields do not interfere with the linear outgoing field in the angular flux, so their energy contribution starts at $\mathcal{O}(\epsilon^2)$.

The immediate numerical extensions are therefore clear: broaden the target-channel scan over (a, ℓ, m, ω), set acceptance gates for the off-diagonal horizon amplitudes, and replace or cross-check the QUADPACK Fourier-tail integral with a dedicated Levin or Filon oscillatory integrator.

Appendix A: Working derivations and normalizations

1. From the scalar wave equation to the radial equation

We use the mostly-plus convention for the Kerr line element. The inverse metric components needed for the scalar wave equation are

$$g^{tt} = -\frac{(r^2+a^2)^2 - a^2\Delta\sin^2\theta}{\Sigma\Delta}, \tag{A1}$$

$$g^{t\phi} = -\frac{2Mar}{\Sigma\Delta}, \tag{A2}$$

$$g^{\phi\phi} = \frac{\Delta - a^2\sin^2\theta}{\Sigma\Delta\sin^2\theta}, \tag{A3}$$

$$g^{rr} = \frac{\Delta}{\Sigma}, \qquad g^{\theta\theta} = \frac{1}{\Sigma}, \qquad \sqrt{-g} = \Sigma\sin\theta. \tag{A4}$$

Thus

$$\Sigma\Box\Phi = \frac{\partial}{\partial r}\left(\Delta\frac{\partial\Phi}{\partial r}\right) + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Phi}{\partial\theta}\right) + \Sigma g^{tt}\frac{\partial^2\Phi}{\partial t^2} + 2\Sigma g^{t\phi}\frac{\partial^2\Phi}{\partial t\,\partial\phi} + \Sigma g^{\phi\phi}\frac{\partial^2\Phi}{\partial\phi^2}. \tag{A5}$$

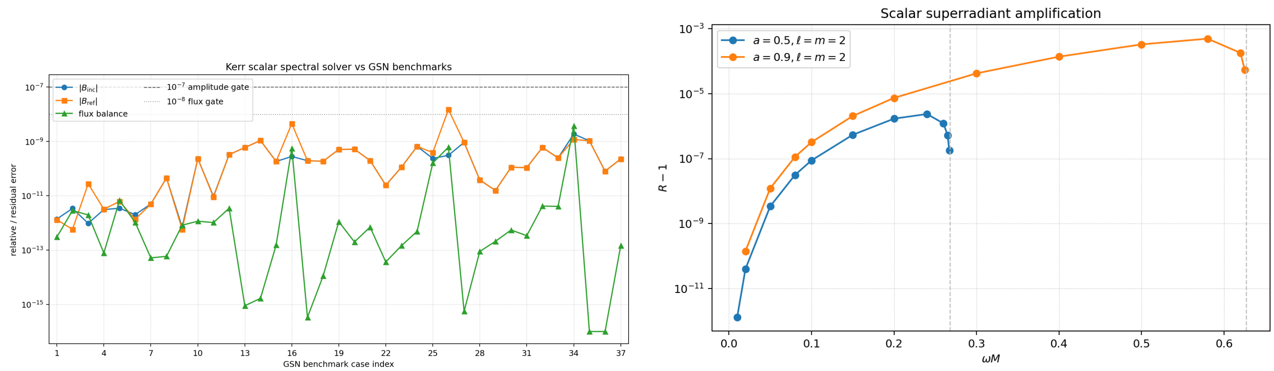

FIG. 1. Linear validation. Left: invariant benchmark errors against GeneralizedSasakiNakamura.jl. Right: frequency sweep showing scalar Kerr superradiant amplification only for $\omega < m\Omega_H$.

TABLE II. Self-channel nonlinear flux coefficients in fixed incident normalization. The source model is Eq. (20); computations use $N = 288$, $q = 224$, and Fourier-tail tolerance 10^{-11}.

case	$M\omega$	$T^{(0)}$	$R^{(0)}$	$T^{(1)}$	$R^{(1)}$
Schwarzschild $\ell = 0$	0.10	3.245×10^{-1}	6.755×10^{-1}	8.726×10^{-2}	-8.726×10^{-2}
Kerr $a = 0.5, \ell = m = 2$ superradiant	0.24	-2.373×10^{-6}	1.000002373	-2.153×10^{-7}	2.039×10^{-7}
Kerr $a = 0.5, \ell = m = 2$ absorbing	0.30	1.595×10^{-5}	9.999840×10^{-1}	1.582×10^{-6}	-1.582×10^{-6}
Kerr $a = 0.9, \ell = m = 2$ superradiant	0.58	-4.970×10^{-4}	1.000497031	-7.819×10^{-5}	7.819×10^{-5}

Substituting $\Phi = R(r)S(\theta)e^{\mathrm{i}m\phi - \mathrm{i}\omega t}$ gives

$$
\begin{aligned}
0 = {} & \frac{1}{R}\frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta\frac{\mathrm{d}R}{\mathrm{d}r}\right) + \frac{1}{S\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}S}{\mathrm{d}\theta}\right) \\
& + \frac{\omega^2\left[(r^2+a^2)^2 - a^2\Delta\sin^2\theta\right]}{\Delta} - \frac{4Mar\omega m}{\Delta} \\
& - \frac{m^2(\Delta - a^2\sin^2\theta)}{\Delta\sin^2\theta}.
\end{aligned} \tag{A6}
$$

Define the frequency-dependent part of Eq. (A6) as

$$
\begin{aligned}
\mathcal{T} = {} & \frac{\omega^2\left[(r^2+a^2)^2 - a^2\Delta\sin^2\theta\right]}{\Delta} - \frac{4Mar\omega m}{\Delta} \\
& - \frac{m^2(\Delta - a^2\sin^2\theta)}{\Delta\sin^2\theta}.
\end{aligned} \tag{A7}
$$

TABLE III. Unit-horizon first-order Green-function amplitudes for the first Kerr target-channel scan.

ℓ'	$\lvert C^{(0)}_{\ell'\ell m}\rvert$	$\lvert A^{(1)}_{\mathrm{ref},\ell'\lvert\ell m}\rvert$	$\lvert A^{(1)}_{H,\ell'\lvert\ell m}\rvert$
2	1.136×10^{-1}	1.137×10^{6}	3.804×10^{3}
3	0	0	0
4	3.578×10^{-2}	1.362×10^{4}	1.319
5	0	0	0
6	5.341×10^{-3}	2.265×10^{3}	5.383×10^{-7}

Using $\Delta = r^2 - 2Mr + a^2$, this can be reorganized as

$$
\begin{aligned}
\mathcal{T} = {} & \frac{\left[(r^2+a^2)\omega - am\right]^2}{\Delta} - a^2\omega^2 + 2am\omega \\
& + a^2\omega^2\cos^2\theta - \frac{m^2}{\sin^2\theta}.
\end{aligned} \tag{A8}
$$

The first line on the right of Eq. (A8) is radial and the second line is angular. Setting the angular part equal to $-A_{\ell m}$ gives the spheroidal equation in the main text. The radial equation then becomes

$$
\frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta\frac{\mathrm{d}R}{\mathrm{d}r}\right) + \left[\frac{K^2}{\Delta} - \left(A_{\ell m} + a^2\omega^2 - 2am\omega\right)\right]R = 0, \tag{A9}
$$

which is Eq. (9) with $\lambda = A_{\ell m} + a^2\omega^2 - 2am\omega$.

2. Asymptotic branches and flux balance

Near the future horizon,

$$
\Delta = (r_+ - r_-)(r - r_+) + \mathcal{O}\left((r - r_+)^2\right), \tag{A10}
$$

$$
K(r_+) = (r_+^2 + a^2)p. \tag{A11}
$$

Since $\Delta\,\mathrm{d}/\mathrm{d}r = (r^2+a^2)\mathrm{d}/\mathrm{d}r_*$, the leading radial equation is

$$
\frac{\mathrm{d}^2R}{\mathrm{d}r_*^2} + p^2R = 0, \qquad r \to r_+. \tag{A12}
$$

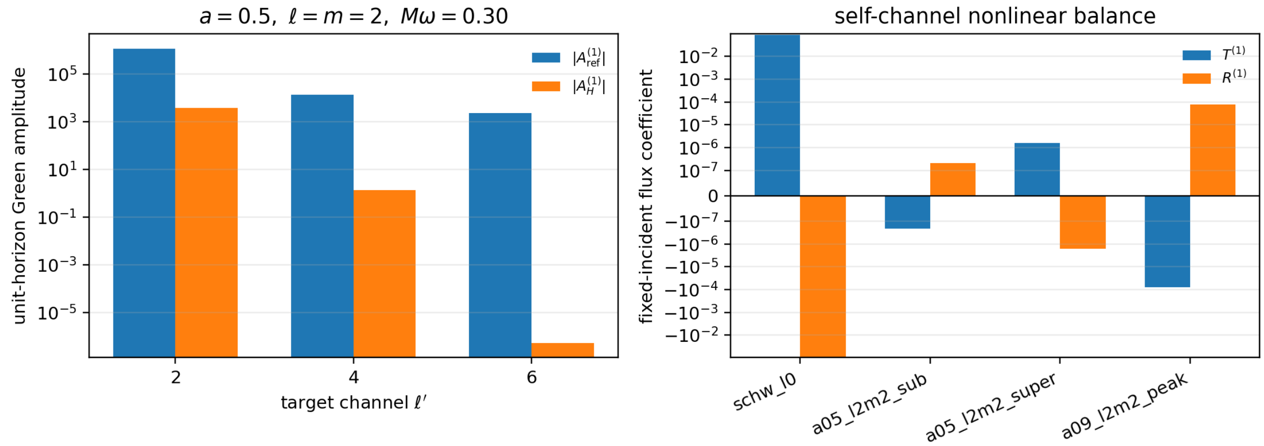

FIG. 2. Nonlinear Kerr scalar results. Left: unit-horizon Green-function amplitudes in the first active target-channel scan. Right: fixed-incident self-channel nonlinear flux coefficients. A symmetric logarithmic scale is used in the right panel to show both superradiant and absorbing cases.

The two local branches are therefore $e^{\mp \mathrm{i}pr_*}$. The in solution uses the future-horizon ingoing branch $e^{-\mathrm{i}pr_*}$. At infinity the same radial equation reduces to

$$\frac{\mathrm{d}^2(rR)}{\mathrm{d}r_*^2} + \omega^2(rR) = \mathcal{O}(r^{-1})\, rR, \tag{A13}$$

which gives the two branches $e^{\mp \mathrm{i}\omega r_*}/r$.

For any homogeneous solution, the radial part of the Klein-Gordon current is

$$J[R] = \frac{1}{2\mathrm{i}}\Delta\left(R^*\frac{\mathrm{d}R}{\mathrm{d}r} - R\frac{\mathrm{d}R^*}{\mathrm{d}r}\right), \qquad \frac{\mathrm{d}J}{\mathrm{d}r} = 0. \tag{A14}$$

For the horizon-normalized in mode,

$$J_H = -p(r_+^2 + a^2), \tag{A15}$$

$$J_\infty = -\omega|B^{\rm inc}|^2 + \omega|B^{\rm ref}|^2. \tag{A16}$$

Equating the two gives Eq. (15). Dividing by the incident energy flux $\omega|B^{\rm inc}|^2$ yields

$$\frac{p(r_+^2 + a^2)}{\omega|B^{\rm inc}|^2} + \left|\frac{B^{\rm ref}}{B^{\rm inc}}\right|^2 = 1. \tag{A17}$$

This also fixes the sign convention for superradiance: if $p < 0$, the horizon energy flux is negative relative to the incident flux and the reflected flux must be larger than the incident flux.

3. Endpoint collocation equations

After the compactification $z = r_+/r$, the horizon is a regular singular endpoint and infinity is an irregular endpoint before phase removal. The phase factors used for the in, down, and up solutions remove precisely the non-analytic oscillatory pieces:

$$R = e^{\mathrm{i}\sigma r_*} r^{-\nu} u(z), \qquad (\sigma, \nu) = (-p, 0), (-\omega, 1), (+\omega, 1), \tag{A18}$$

for in, down, and up solutions respectively. The remaining function has a regular endpoint expansion,

$$u(z) = u_0 + u_1(z - z_e) + u_2(z - z_e)^2 + \cdots, \qquad z_e = 0 \text{ or } 1. \tag{A19}$$

Inserting this expansion into Eq. (27) gives

$$B_1(z_e)u_1 + B_0(z_e)u_0 = 0, \qquad B_2(z_e) = 0. \tag{A20}$$

This is the boundary equation used in the first or last collocation row. One additional row sets the normalization $u_0 = 1$. No artificial horizon cutoff is introduced; the endpoint itself is part of the spectral grid.

4. Green function and nonlinear amplitudes

Let $y_1 = R^{\rm in}_{\ell'}$ and $y_2 = R^{\rm up}_{\ell'}$ solve the target homogeneous equation. For the Sturm-Liouville form

$$\mathcal{L}_{\ell'} y = \frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta\frac{\mathrm{d}y}{\mathrm{d}r}\right) + V_{\ell'}(r)y, \tag{A21}$$

the Wronskian

$$\mathcal{W}_{\ell'} = \Delta(y_1 y_2' - y_2 y_1') \tag{A22}$$

is constant. The Green function

$$G(r, r') = \frac{y_1(r_<)y_2(r_>)}{\mathcal{W}_{\ell'}} \tag{A23}$$

is continuous at $r = r'$ and satisfies

$$\Delta\left[\frac{\partial G}{\partial r}\right]_{r=r'^-}^{r=r'^+} = 1. \tag{A24}$$

Thus $\mathcal{L}_{\ell'} G = \delta(r - r')$. Since the nonlinear radial equation is $\mathcal{L}_{\ell'} R^{(1)}_{\ell'} = -Q_{\ell'}$, the solution satisfying outgoing radiation at infinity and ingoing radiation at the future horizon is

$$R^{(1)}_{\ell'}(r) = -\int_{r_+}^{\infty} G(r, r') Q_{\ell'}(r')\, \mathrm{d}r'. \tag{A25}$$

Taking $r \to \infty$ gives

$$R^{(1)}_{\ell'} \sim A^{(1)}_{\mathrm{ref},\ell'|\ell m} \frac{e^{+\mathrm{i}\omega r_*}}{r}, \tag{A26}$$

$$A^{(1)}_{\mathrm{ref},\ell'|\ell m} = -\frac{1}{\mathcal{W}_{\ell'}} \int_{r_+}^{\infty} y_1 Q_{\ell'}\, \mathrm{d}r, \tag{A27}$$

and taking $r \to r_+$ gives

$$R^{(1)}_{\ell'} \sim A^{(1)}_{H,\ell'|\ell m} e^{-\mathrm{i}p_{\ell'} r_*}, \tag{A28}$$

$$A^{(1)}_{H,\ell'|\ell m} = -\frac{1}{\mathcal{W}_{\ell'}} \int_{r_+}^{\infty} y_2 Q_{\ell'}\, \mathrm{d}r. \tag{A29}$$

These are the formulas used by the production scripts.

5. Why the first-order self-channel balance is expected

For the self channel, the source has the form

$$Q(r) = F(r) |R_{\mathrm{in}}|^2 R_{\mathrm{in}}, \qquad F(r) = r^2 C^{(0)}_{\ell\ell m} + a^2 C^{(2)}_{\ell\ell m}, \tag{A30}$$

where $F(r)$ is real for the present real-frequency spheroidal basis. Let R_1 solve $\mathcal{L} R_1 = -Q$. Combining this equation with the complex conjugate homogeneous equation for R^*_{in} gives

$$\frac{\mathrm{d}}{\mathrm{d}r} \left[\Delta \left(R^*_{\mathrm{in}} R_1' - R_1 R^{*\prime}_{\mathrm{in}} \right) \right] = -R^*_{\mathrm{in}} Q = -F(r) |R_{\mathrm{in}}|^4. \tag{A31}$$

The right-hand side is real. Therefore the imaginary part of the bracket is constant between the horizon and infinity. Evaluating that constant using the asymptotic amplitudes of R_{in} and R_1 gives precisely

$$\frac{2p(r_+^2 + a^2)}{\omega |B^{\mathrm{inc}}|^4} \mathrm{Re}\, A^{(1)}_H + \frac{2\,\mathrm{Re}\left[\left(B^{\mathrm{ref}}\right)^* A^{(1)}_{\mathrm{ref}} \right]}{|B^{\mathrm{inc}}|^4} = 0, \tag{A32}$$

which is Eq. (40). The numerical residual in Table II is therefore a direct measure of the combined spectral, matching, and oscillatory-quadrature error.

Appendix B: Schwarzschild limit

For $a = 0$, $r_+ = 2M$, $\Omega_H = 0$, $S_{\ell m}$ reduces to $Y_{\ell m}$, and Eq. (9) is equivalent to the Regge–Wheeler scalar equation for the master variable $\psi = rR$:

$$\left[\frac{\mathrm{d}^2}{\mathrm{d}r_*^2} + \omega^2 - f \left(\frac{\ell(\ell+1)}{r^2} + \frac{2M}{r^3} \right) \right] \psi = 0, \qquad f = 1 - \frac{2M}{r}. \tag{B1}$$

The source Eq. (20) becomes $r^2 C^{(0)}_{\ell'\ell m} |R|^2 R$. In the master variable this is the familiar r^{-2} cubic radial weight of the Schwarzschild nonlinear Green-function problem. Thus the Kerr implementation reduces to the base method when $a = 0$, while retaining the endpoint-collocation boundary treatment.

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